Local-first AI course assistant
Project Proposal

Local-First Course AI Packs for LM Studio

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1. Application Overview

SP2 is a local course-assistance system that helps students receive AI answers grounded in material selected by their teacher. A teacher gives SP2 a supported course document, and SP2 converts it into a portable course pack. A student installs that pack and asks questions through LM Studio. SP2 retrieves relevant course content, while LM Studio remains responsible for the chat interface and final answer.

The application is local-first: course packs, metadata, vectors, and retrieval data are stored on the user's computer. SP2 does not replace LM Studio or an LMS. It provides the ingestion and retrieval layer between teacher material and the AI chat.

System Structure

Part	Responsibility
Teacher side	Accept PDF, ODT, or DOCX files; extract text; create chunks and embeddings; export a course-pack ZIP.
Student side	Validate and install packs; list or delete installed packs; retrieve course context for questions.
Storage	Keep installed files locally, pack metadata in SQLite, and searchable chunks and vectors in LanceDB.
MCP server	Connect LM Studio to the SP2 FastAPI backend and expose ingestion, pack-management, and retrieval tools.

2. Main Functional Focus

SP2 can be described using two main functions:

1. **Ingestion:** teacher document → validated and searchable installed course pack.
2. **Retrieval:** selected pack + student question → relevant course context for LM Studio.

The two functions form one continuous workflow:

Teacher material → SP2 ingestion → local course pack → SP2 retrieval → LM Studio answer

3. Functional Specification — Ingestion

Purpose

Ingestion converts a teacher's source document into a portable course pack and installs it into the student's local SP2 storage.

Item	Specification
Actor	Teacher, directly or through LM Studio
Input	Local path to a PDF, ODT, or DOCX file
Preconditions	SP2 backend is running; source file exists; LM Studio embedding model is available
Output	Installed pack ID, pack metadata, chunk count, ZIP path, and searchable local pack

Functional Behavior

1. SP2 validates that the path is non-empty, exists, is a file, and has a supported extension.
2. The teacher service selects the PDF, ODT, or DOCX extractor and converts the source into normalized text.
3. SP2 divides the text into overlapping chunks and keeps source details such as title, page, section, and chunk index.
4. The chunks are sent to LM Studio's local embedding endpoint using **text-embedding-nomic-embed-text-v1.5**.
5. SP2 writes **pack.json**, **chunks.json**, and **vectors.npy**, then exports them as a ZIP.
6. The ZIP is validated and installed: files go to the installed-packs directory, metadata goes to SQLite, and chunks and vectors go to LanceDB.

Main Validation and Failure Rules

- The pack must contain exactly the three required files with valid metadata and unique chunk IDs.
- The number of vectors must equal the number of chunks, and the vector dimension must match the metadata.
- Missing, empty, malformed, or unsupported files are rejected with an explanatory error.
- If installation fails, SP2 attempts to remove the new files, SQLite record, and LanceDB rows.
- A duplicate **{pack_id}-{version}** installation is rejected because replacement behavior is not yet implemented.

Successful Result

The MCP workflow returns mode **course_pack_ready**, the local pack number, title, chunk count, storage paths, and embedding information. That pack number can then be used for retrieval.

4. Functional Specification — Retrieval

Purpose

Retrieval finds course content related to a student's question from one selected installed pack. SP2 returns evidence as structured context; LM Studio writes the final answer.

Item	Specification
Actor	Student using LM Studio
Inputs	Positive installed-pack ID and non-empty question
Preconditions	Pack is installed and active; SP2 and LM Studio embedding services are running
Output	A **course_context** or **no_course_context** packet

Functional Behavior

1. LM Studio calls **sp2_get_course_context(pack, question)**.
2. The MCP adapter validates the pack number and normalizes the question.
3. SP2 confirms that the installed pack exists, is active, and uses the supported embedding model.
4. The question is embedded using the same model and dimension used to build the pack.
5. LanceDB searches for the nearest chunks while filtering by **installed_pack_id**, so results cannot come from another pack.
6. SP2 returns the selected chunks with their source information, text, distance, and calculated relevance score.
7. LM Studio uses the returned context to prepare the final response for the student.

Main Validation and Failure Rules

- The pack number must be a positive integer and the question cannot be empty.
- Missing or inactive packs are rejected.
- Model or vector-dimension mismatches stop retrieval.
- LM Studio connection failures, timeouts, and malformed embedding responses produce a clear service error.
- If no chunk passes an optional distance limit, SP2 returns a valid **no_course_context** packet instead of inventing course evidence.

Successful Result

For a match, SP2 returns mode **course_context**, the selected pack information, normalized question, embedding model, and relevant chunks. Retrieval is read-only and does not change the installed pack.

The backend already supports optional **top_k** and **max_distance** controls. The current MCP tool exposes only **pack** and **question**; exposing a relevance limit through MCP is a suitable future improvement.